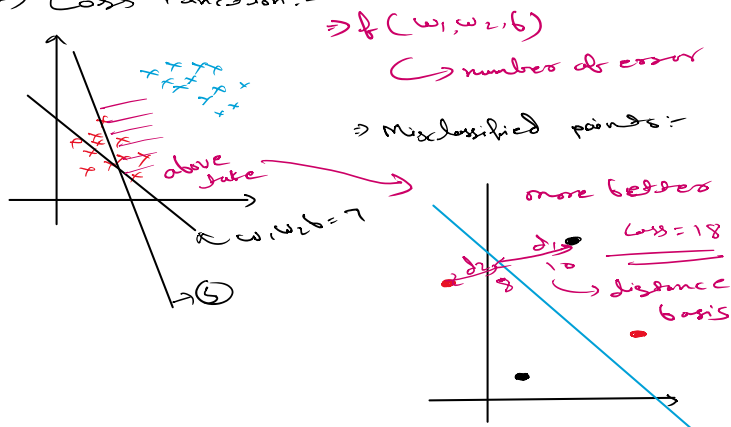
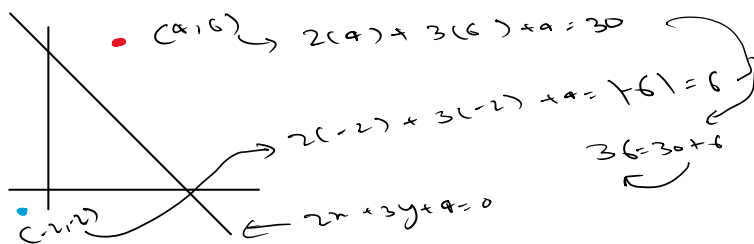


⇒ Loss Function:-



⇒ Actual loss function used in the perceptron.



⇒ this method used over distance because it is easy to calculate as compare to distance.

$$\Rightarrow L(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \alpha R(w_1, w_2)$$

↓
Regularization

$$* L(y_i, f(x_i)) = \max(0, -y_i f(x_i))$$

↓
 $f(x_i) = w_1 x_1 + w_2 x_2 + b$

$$L = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

n: # data in data
 w_1, w_2, b
 L: argument
 w_1, w_2, b
 gradient descent

⇒ Explanation of loss function:-

$$L = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

x_1	x_2	y
x_{11}	x_{12}	y_1
x_{21}	x_{22}	y_2

where $f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$

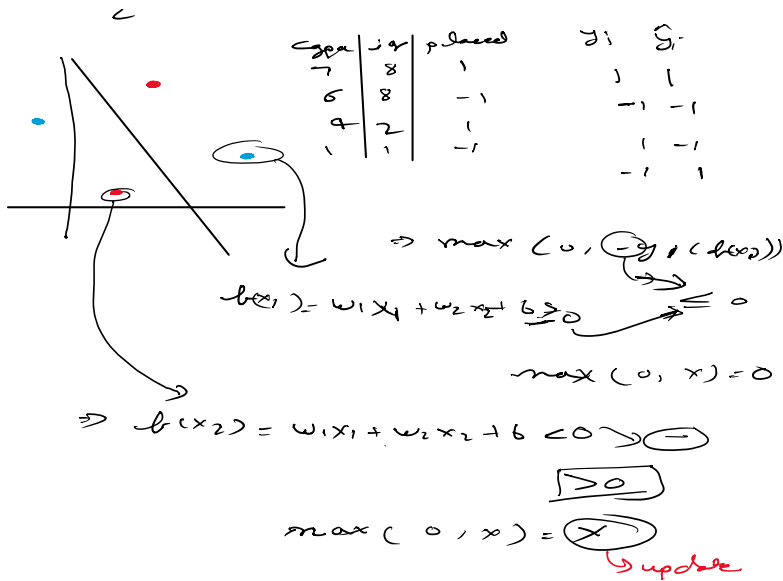
$$\Rightarrow \max(0, -y_i f(x_i)) = \max(0, x)$$

↳ $\begin{matrix} \geq 0 = x \\ \leq 0 = 0 \end{matrix}$

⇒ Now we learn with one help of example:-

$$L = \frac{1}{2} [\max(0, -y_1 f(x_1)) + \max(0, -y_2 f(x_2))] = 0$$

input	x_1	x_2	output
1	8	1	1
0	8	-1	-1



$\Rightarrow$ Gradient Descent:-

$$L = \arg \min_{w_1, w_2, b} \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

$$L = L(w_1, w_2, b)$$

$$\text{where } f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

$\Rightarrow$ algo:-

for i in epochs:

$$w_1 = w_1 + \eta \frac{\partial L}{\partial w_1} \quad \left[\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b} \right]$$

$$w_2 = w_2 + \eta \frac{\partial L}{\partial w_2}$$

$$b = b + \eta \frac{\partial L}{\partial b}$$

$\Rightarrow$ loss function differentiation:-

$$L = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

$$f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

$$* \frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f(x_i)} \times \frac{\partial f(x_i)}{\partial w_1}$$

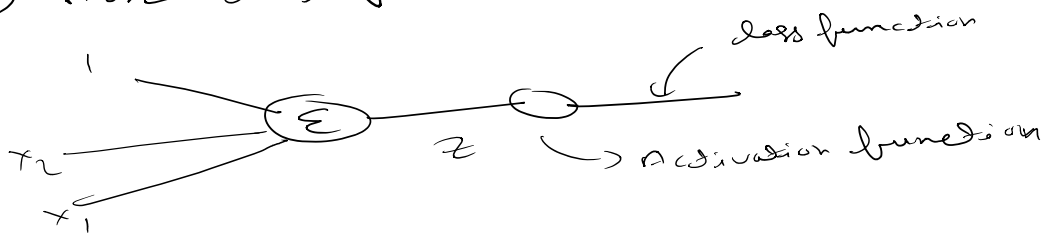
$$\frac{\partial L}{\partial f(x_i)} = \begin{cases} 0, & \text{if } y_i f(x_i) \geq 0 \\ -y_i, & \text{if } y_i f(x_i) < 0 \end{cases} \quad \frac{\partial f(x_i)}{\partial w_1} = x_{i1}$$

$$\frac{\partial L}{\partial w_1} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i1} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\frac{\partial L}{\partial w_2} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i2} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\frac{\partial L}{\partial b} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i & \text{if } y_i f(x_i) < 0 \end{cases}$$

⇒ more loss function: -



⇒ Indesible: -

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

① Loss function: -

⇓
binary cross entropy

$$L = -y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)$$

⇒ Perceptron == Logistic Regression

② multiclass

Activation

↳ softmax

Cross function

⇓
categorical cross entropy

$$L = \sum_{i=1}^N y_i \log(\hat{y}_i)$$

⇒ ① Loss function
Hinge loss

② (log loss
(binary cross entropy))

③ Categorical cross entropy

④ mse

Activation
Step

Sigmoid

Softmax

Linear

output

perceptron →
binary classified
(-1, 1)

logistic
regression

0-1 → binary

softmax →
regression probability
multiclass
classified

Linear Regression